GatedMemorySAM

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Method. We introduce GatedMemorySAM, which builds upon MemorySAM [1] that repurposes SAM2's [2] temporal memory attention for cross-modal fusion by treating each input modality (RGB, LiDAR, Thermal) as a separate "frame" in SAM2's video pipeline. We extend this framework with two key modifications: (1) Soft MoE LoRA adaptation, and (2) quality-aware modality scoring.

For backbone adaptation, we replace standard LoRA [3] with Soft MoE routing [4] over multiple LoRA experts, inserted into every attention block (Q and V projections, 48 layers total) with rank=4 and 3 experts. Each spatial token is softly routed to all experts via a learned gating network, enabling per-token specialization across modalities.

For modality weighting, we introduce a CrossModalFusionHead that compares all modality features via global average pooling and a shared comparison MLP to produce per-modality softmax weights. These weights drive two score-based fusion mechanisms: (i) memory modulation, which max-normalizes the scores and scales each modality's backbone features before they enter SAM2's memory bank, so that the best modality retains full strength while weaker ones are suppressed; and (ii) weighted mask fusion, which averages the per-modality decoder outputs using the same softmax weights. The overall pipeline is illustrated in Figure 1.

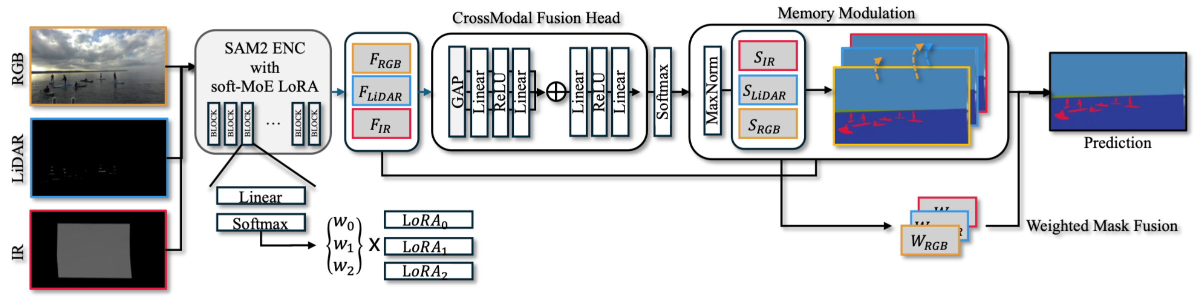

Figure 1: Overview of GatedMemorySAM. Each modality is encoded by SAM2 with Soft-MoE LoRA, scored by the CrossModalFusionHead, and fused via memory modulation and weighted mask fusion.

Training Details. We train with the AdamW optimizer (lr=6e-4, weight decay=0.01) using a warmup polynomial LR schedule (10-epoch warmup, power=0.9) for 200 epochs. The loss function is Online Hard Example Mining Cross-Entropy (OHEM-CE). Input images are resized to 1024×1024. We use DDP training with an effective batch size of 16 across 8 GPUs with gradient accumulation.

Night Augmentation. Since the test set consists entirely of nighttime images while training data is daytime, we apply an extensive nighttime simulation pipeline: brightness darkening (range [0.03, 0.45] with 60% dark-biased sampling), contrast reduction, gamma correction ([0.4, 0.8]), Gaussian and Poisson shot noise, cross-modal replacement (CRM, p=0.20), and PhysAug [5]-inspired spatial perturbations (random convolution filters and planar Fourier wave patterns, p=0.40).

Datasets and Pretrained Weights. We train exclusively on the MULTIAQUA dataset provided by the challenge organizers. The SAM2 Hiera-B+ backbone is initialized from publicly available pretrained weights. No additional external datasets or annotations are used.

Hardware and FPS. Training was conducted on 8× NVIDIA RTX 3090 GPUs (24GB each). Inference runs on a single NVIDIA Titan RTX GPU at approximately 2.1 FPS (1024×1024 input, 3 modalities processed sequentially).

Maritime Domain Adaptations. The primary challenge is the extreme day-night domain gap (daytime training vs. nighttime-only test). Our nighttime simulation augmentation was the single most impactful component, nearly doubling the test mIoU.

Submission. Our best submission (Submission #16710, leaderboard name: *kblkblkbl*) achieved a validation mIoU of 93.29% and a test mIoU of 70.91%, yielding an M-score of 82.10, where M $= 0.75\times$ $\mathrm{mIoU}_{\mathrm{val}} + 0.25\times$ $\mathrm{mIoU}_{\mathrm{test}}$.

References

[1] C. Liao, X. Zheng, Y. Lyu, H. Xue, Y. Cao, J. Wang, K. Yang, and X. Hu, "MemorySAM: Memorize modalities and semantics with segment anything model 2 for multi-modal semantic segmentation," *arXiv preprint arXiv:2503.06700*, 2025.

[2] N. Ravi, V. Gabber, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolber, L. Gustafson, E. Mintun, J. Pan, K. V. Alwala, N. Carion, C.-Y. Wu, R. Girshick, P. Dollár, and C. Feichtenhofer, "SAM 2: Segment anything in images and videos," *arXiv preprint arXiv:2408.00714*, 2024.

[3] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations (ICLR)*, 2022.

[4] J. Puigcerver, C. Riquelme, B. Mustafa, and N. Houlsby, "From sparse to soft mixtures of experts," in *International Conference on Learning Representations (ICLR)*, 2024.

[5] X. Xu, J. Yang, W. Shi, S. Ding, L. Luo, and J. Liu, "PhysAug: A physical-guided and frequency-based data augmentation for single-domain generalized object detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, pp. 21815–21823, 2025.